Managing Aggregate Demand with Consumption Taxes

Axelle Ferriere[1,2] **Gaston Navarro**[3] **Morgane Richard**[1]

[1]Sciences Po [2]CNRS, CEPR [3]Federal Reserve Bank of Richmond

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    Dot-com Recession: Defense spending, Economic Growth & Tax Relief Reconciliation Act (2001)

    Great Recession: Economic Stimulus Act (2008), American Recovery and Reinvestment Act (2009)

    Covid Recession: Coronavirus Aid, Relief, and Economic Security (2020)

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- Alternative fiscal instrument to (better) stabilize through private consumption?

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- Fiscal burden contingent on effectiveness of the stimulus

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  - Standard HANK model with unemployment
  - Multipliers of consumption tax cuts vs. spending shocks vs. transfers

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- Theoretical equivalence in the simplest possible setup
- Quantitative evaluation of consumption tax cuts
  - Standard HANK model with unemployment
  - Multipliers of consumption tax cuts vs. spending shocks vs. transfers
- Effectiveness to stabilize demand-driven recessions
  - Optimal systematic rule with the business cycle

Literature

- Effects of monetary policy and government spending in HANK models

  Kaplan, Moll, and Violante (2018), Hagedorn, Manovskii, and Mitman (2019), Bilbiie (2020), Auclert, Rognlie, and Straub (2023), Ferriere and Navarro (2024), Alves and Violante (2023)

- Optimal fiscal and monetary policy in HANK

  Bhandari, Evans, Golosov, and Sargent (2021), Le Grand and Ragot (2024), McKay and Wolf (2023)
  Broer, Druedahl, Harmenberg, and Oberg (2025), Le Grand, Ragot and Bourany (2025)

- Stabilization and consumption taxes in HANK

  Parodi (2024), Bachmann, Born, Goldfayn-Frank, Kocharkov, Luetticke, & Weber (2025), Bartal & Becard (2025)

- Equivalence results in HANK

  Correia, Nicolini & Teles (2008), Correia, Farhi, Nicolini & Teles (2013), Seidl & Seyrich (2023), Wolf (2024), Wolf (2025)

- Evidence on consumption tax cuts

  Blundell (2009), Benzarti, Carloni, Harju, & Kosonen (2020), Bachmann et al. (2025)

A Theoretical Equivalence

The Simplest Model

- RBC model with no capital, no heterogeneity
- Representative household
  - Values consumption C_t and leisure $1 - N_t$
  - Can save in a risk-free bond A_t
- Representative competitive firm
  - Hires labor in a competitive labor market to produce output Y_t
- Government has to finance public good G_t
  - Uses labor and consumption taxes τ_t^n and τ_t^c
  - Issues public debt $B_t = A_t$ in equilibrium
- Prices: (r_t, w_t) the interest rate and the net-of-taxes wage

Two Expansions

- From steady state with $\tau_{ss}^c = 0$ (wlog), consider two temporary fiscal expansions.

1. Government spending (expansion) stimulus $\left\{\hat{G}_t\right\}_{t=0}^{T}$
   - Fiscal cost $\hat{F}_t \equiv \hat{G}_t - G$
   - Financed with any policy $\hat{\Pi}_t \equiv \left(\hat{\tau}_t^n, \hat{B}_t\right)$ converging back to steady state
   - Output & after-tax prices $\hat{X}_t \equiv \left(\hat{Y}_t, \hat{r}_t, \hat{w}_t\right)$ and consumption $\hat{C}_t$

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2. Consumption tax (cut) stimulus $\left\{\bar{\tau}_t^c < \tau_{ss}^c = 0\right\}_{t=0}^{T}$
   - Fiscal cost $\bar{F}_t \equiv -\bar{\tau}_t^c \bar{C}_t$ with policy $\bar{\Pi}_t \equiv \left(\bar{\tau}_t^n, \bar{B}_t\right)$
   - Output & after-tax prices $\bar{X}_t \equiv \left(\bar{Y}_t, \bar{r}_t, \bar{w}_t\right)$ and consumption $\bar{C}_t$

Equivalence Result

- **Assumption:** Assume log-separable preferences $U(c, n) = \log c - v(n)$.
- **Proposition:** Let the two policies be identical $\bar{\Pi}_t = \bar{\Pi}_t \ \forall t$. Then,
  - The equilibrium $\bar{\tau}_t^c$ is such that $\hat{F}_t = \bar{F}_t \ \forall t$;
  - Output and after-tax prices are identical: $\bar{X}_t = \bar{X}_t \ \forall t$;
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    - The consumption tax cut that replicates the spending stimulus is given by:

$$\bar{\tau}_t^c = -\frac{\hat{G}_t - G}{\hat{Y}_t - G}$$

Intuition The log case

- A recursive formulation of the household problem reads:

$$V_t(A_t) = \max_{C_t, N_t, A_{t+1}} \{\log(C_t) - v(N_t) + \beta V_{t+1}(A_{t+1})\} \text{ s.t.}$$

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  - Household decisions for $\{N_t, A_{t+1}\}$ independent of $\bar{\tau}_t^c$
  - Consistent with firms decisions
  - Consistent with government's budget constraint as identical cost
  - Consistent with market clearing if and only if $\bar{C}_t = \hat{C}_t + \left(\hat{G}_t - G\right) = \left(\hat{Y}_t - G\right)$

Extending the Equivalence

- Works with aggregate shocks and non-zero consumption tax
  - Let X_t denote a variable in t in absence of fiscal stimulus;
  - Fiscal costs defined as:

    $\hat{F}_t \equiv (\hat{G}_t - G_t) - \tau_t^c(\hat{C}_t - C_t)$ and $\bar{F}_t \equiv -(\bar{\tau}_t^c - \tau_t^c)\bar{C}_t - \tau_t^c(\bar{C}_t - C_t)$
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- **Equivalence** of any $\{\tilde{G}_t, \tilde{\tau}_t^c\}$ that deliver the same fiscal cost:
  $\tilde{F}_t \equiv (\tilde{G}_t - G_t) - (\tilde{\tau}_t^c - \tau_t^c)\tilde{C}_t - \tau_t^c(\tilde{C}_t - C_t)$
  - Finance spending shock with consumption taxes $\Rightarrow$ multiplier $= 0$
  - Full crowding-out of private consumption!

Robustness

- Robust to several model extensions
  - Capital in the production function
  - Heterogeneous agents
  - Various labor supply arrangements (extensive margin)
  - Nominal rigidities: sticky prices, sticky wages

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- Key assumption: **log preferences**
  - Two key margins: static (labor supply), dynamic (savings decisions)
  - Formula away from log case? Bound multipliers with sufficient statistics
    - Static responses: wealth effects, MPE, labor supply elasticity (lotteries, . . .)
    - Dynamic response: IES, response of savings to changes in interest rates (UK mortgages, . . .)

Literature

- Correia, Nicolini & Teles (2008), Correia, Farhi, Nicolini & Teles (2013)
  - Replicate monetary policy by a combination of $\{\tau_t^c, \tau_t^n, \tau_t^k\}$ in a RANK
  - Seidl & Seyrich (2023): Extension in HANK
    - Requires additional adjustment in public debt; Holds in the cross-section
  - Wolf (2025): Equivalence of monetary policy with transfers
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  - Conditions under which changes in private spending $\Leftrightarrow$ changes in public spending
    - Same path for aggregate demand
    - Same path of future taxes

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  - Conditions under which changes in private spending $\Leftrightarrow$ changes in public spending
    - Same path for aggregate demand
    - Same path of future taxes
  - Application: $G \Leftrightarrow T$
    - Same labor supply response: no wealth effects, or fully demand-determined employment
    - Same fiscal path: fully deficit-financed

Environment

A HANK Model with Unemployment

- Households
  - Heterogeneous in income and wealth
  - Heterogeneous MPCs
- Unemployment risk and Okun's law
  - Unequal incidence across households and increasing in recessions
- Nominal rigidities
  - Aggregate demand effects

Model Overview

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  - Monetary authority implements a standard Taylor rule
- Government
  - Finances spending, transfers, UI benefits with debt, and labor, consumption, capital taxes

Households **Working households**

- Individual state: asset a, discount factor β, productivity x, and employment $\eta \in \{\ell, u\}$

- Value function when employment "island" $\eta = \ell$

$$V_t(a, x, \ell, \beta; \mu_t) = \max_{c,a'} \left\{ \log c - B\frac{h^{1+\varphi}}{1+\varphi} + \beta \mathbb{E}_t \left[V_{t+1}(a', x', \eta', \beta'; \mu_{t+1}) | x, \beta, \ell \right] \right\} \quad \text{s.t.}$$

$$(1+\tau_t^c)c + a' = a + y^\ell + y^k - \mathcal{T}_t(y^\ell, y^k) + T_t,$$

$$y^\ell = w_t x h, \quad y^k = r_t a, \quad a' \geq 0, \quad \mu_{t+1} = \Psi(\mu_t).$$

  - AR(1) process for discount factor, productivity and employment status
  - Flat capital and consumption taxes τ^k and τ_t^c, progressive loglinear labor tax (λ_t, τ^ℓ)
    Heathcote, Storesletten, and Violante (2017)

Households **Unemployed households**

- Value function when in unemployment "island" $\eta = u$

$$V_t(a, x, u, \beta; \mu_t) = \max_{c, a'} \{\log c + \beta \mathbb{E}_t \left[V_{t+1}(a', x', \eta', \beta'; \mu_{t+1}) | x, \beta, u)\right]\} \quad \text{s.t.}$$

$$(1 + \tau_t^c)c + a' = a + y^k + \mathcal{B}_t(w_t x h) - \mathcal{T}_t(0, y^k) + T_t,$$

$$y^k = r_t a, \quad a' \geq 0.$$

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  - Unemployment benefits function of hourly wage

    Kekre (2022)

$$\mathcal{B}_t(w_t x h) = \zeta \min\left(\mathcal{R} w_t x h, \overline{ui}\right) + \chi w_t x h$$

    + ζ to match fraction of recipients, $\mathcal{R}$ the replacement rate, $\overline{ui}$ the UI cap
    + χ to capture household labor income received while in unemployment

  - AR(1) process for discount factors, productivity and employment status

Firms and Labor Market

- Standard two-layer structure
  Hagedorn, Manovskii, and Mitman (2019)

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- Representative final good producer

- Intermediate good producers indexed by k
  - Linear technology: $y_{kt} = n_{kt}$, flexible prices $\rightarrow$ constant wage w

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- Unions indexed by j
  - Demand same labor from all working households and aggregate it linearly
  - Nominal rigidities a la Rotemberg

Unions **Phillips Curve**

- Continuum of unions j demanding labor h_{ji} from households i
  - Labor demand: $n_{jt} = \left(\frac{W_{jt}}{W_t}\right)^{-\epsilon} Y_t$
  - Nominal wage W_{jt} s. to utility cost: $\frac{\Theta}{2}\left(\frac{W_{jt}}{W_{jt-1}} - \bar{\Pi}\right)^2 Y_t$
  - Atomistic union: take as given total consumption and total hours of working households

$$J_t(W_{tj-1}) = \max_{W_{jt}} \left\{ \int \left[\frac{W_{jt}}{P_t} \frac{(1 - MTR_i)x_i}{1 + \tau_{ct}} - \frac{v'(h_i)}{u'(c_i)} \right] h_j di - \frac{\Theta}{2}\left(\frac{W_{jt}}{W_{jt-1}} - \bar{\Pi}\right)^2 Y_t + \beta J_{t+1}(W_{tj}) \right\}$$

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- Continuum of unions j demanding labor h_{ji} from households i
  - Labor demand: $n_{jt} = \left(\frac{W_{jt}}{W_t}\right)^{-\epsilon} Y_t$
  - Nominal wage W_{jt} s. to utility cost: $\frac{\Theta}{2}\left(\frac{W_{jt}}{W_{jt-1}} - \bar{\Pi}\right)^2 Y_t$
  - Atomistic union: take as given total consumption and total hours of working households

$$J_t(W_{tj-1}) = \max_{W_{jt}} \left\{ \int \left[\frac{W_{jt}}{P_t} \frac{(1 - MTR_i)x_i}{1 + \tau_{ct}} - \frac{v'(h_i)}{u'(c_i)} \right] h_j di - \frac{\Theta}{2}\left(\frac{W_{jt}}{W_{jt-1}} - \bar{\Pi}\right)^2 Y_t + \beta J_{t+1}(W_{tj}) \right\}$$

- **Phillips curve** collapses to Erceg et al (2000) with no heterogeneity

$$\left(\Pi_t^w - \bar{\Pi}\right)\Pi_t^w + \frac{\epsilon - 1}{\Theta}\frac{w}{\bar{x}_t}\int \frac{(1 - MTR_i)x_i}{1 + \tau_{ct}} di = \frac{\epsilon}{\Theta}\frac{1}{\bar{x}_t}\int \frac{v'(h)}{u'(c_i)} di + \left(\Pi_{t+1}^w - \bar{\Pi}\right)\Pi_{t+1}^w \frac{Y_{t+1}}{Y_t}$$

Government

- Monetary authority follows a Taylor rule: $1 + i_t = (1 + \bar{i}) \left(\frac{\Pi_t}{\bar{\Pi}}\right)^{\phi_\Pi}$

- Fiscal authority faces a standard borrowing constraint

$$G_t + (1 + r_t)D_t + T_t + \int \mathcal{B}_t(w_t x) d\mu_t = D_{t+1} + \int \mathcal{T}_t(y_t^\ell, y_t^k) d\mu_t$$

- Fiscal rule with parameter Φ_D for public debt, λ_t clears the budget constraint
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  Uhlig (2010)
  - $\Phi_D = 0$ for constant debt, all adjustment in tax level
  - $\Phi_D \to 1$ for constant taxes, all adjustment in debt

    $D_{t+1} = (1 - \phi_D)D + \phi_D \left(\hat{G}_t - \tau^k r_t A_t - \tau_t^c C_t - \mathcal{R}_t^\ell\right)$, where
    + $\hat{G}_t$ denotes total spending, including debt repayments
    + $\mathcal{R}_t^\ell$ denotes fiscal revenues at steady-state labor tax schedule

Steady State **Households**

- Quarterly model calibrated to liquid wealth

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- Stochastic $\beta \in \{\bar{\beta} - \Delta,\ \bar{\beta},\ \bar{\beta} + \Delta\}$, duration of 50 years
  Krusell and Smith (1998)
  - $\bar{\beta}$ s.t. $r \equiv 3.5\%$ annually
  - Δ s.t. top-quintile liquid wealth $\approx 85\%$ (SCF)
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- Labor supply
  - B to match hours $h = 1/3$, Frisch $\varphi^{-1} = 0.4$

- Productivity $(\rho_x, \sigma_x) = (0.989, 0.287)$
  Chang and Kim (2007)

Steady State **Firm and government**

- Technology: $\varepsilon = 7$, $\Theta = 200 \rightsquigarrow$ Phillips curve slope $\varepsilon/\Theta = 0.035$
  Galí and Gertler (1999)

- Government
  - Capital tax $\tau^k = 35\%$, labor income tax progressivity $\tau^\ell = 0.1$, consumption tax $\tau^c = 5\%$
    Chen, Imrohoroglu, and Imrohoroglu (2007), Heathcote, Storesletten, and Violante (2017), Ferriere, Grübener, Navarro, and Vardishvili (2023)
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  - Automatic responses of inflation and debt: $\Phi_\Pi = 1.5$, $\Phi_D = 0.75$

Unemployment Steady State and Business Cycles

- Job finding rates and separation rates across hourly wage distribution

  Mueller (2017)

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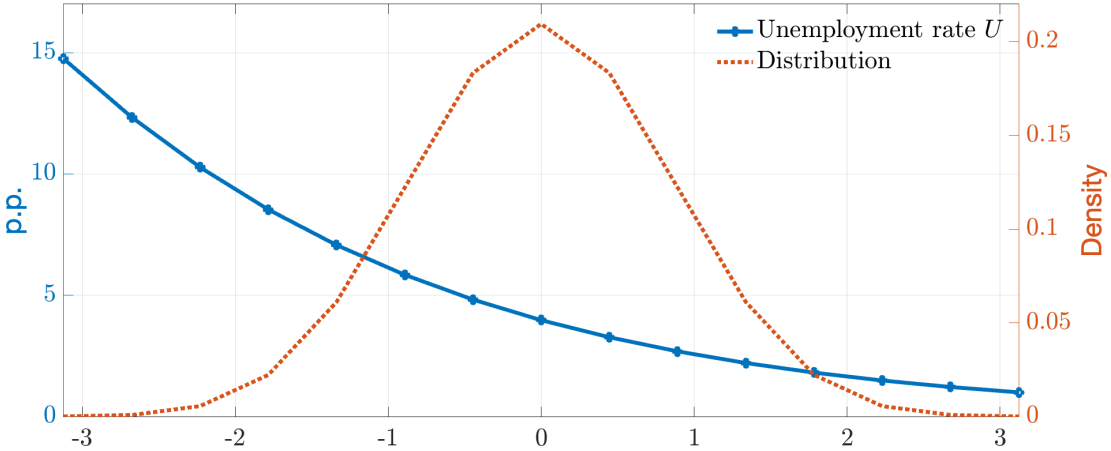
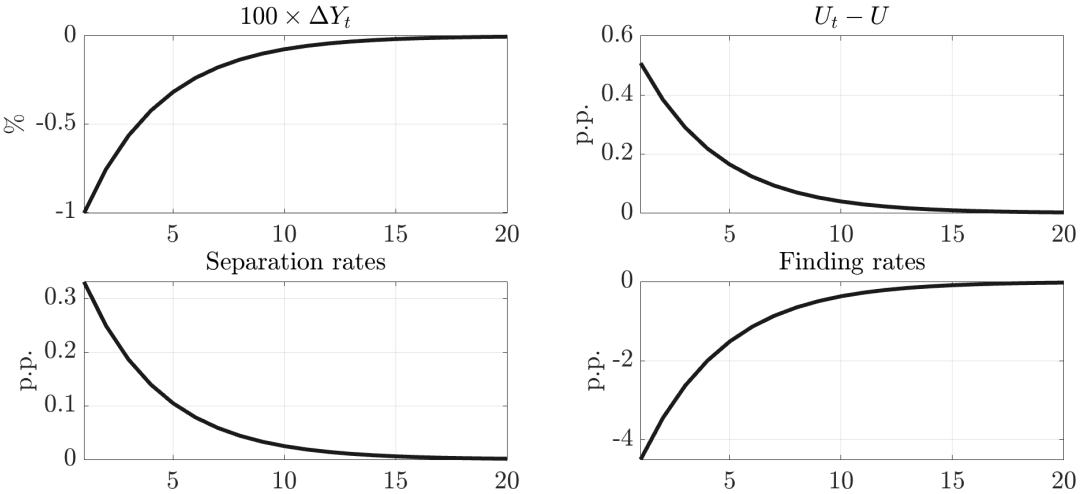
- Okun's law: Okun coefficient $c_{OK} = 0.5$
  Ball, Leigh, and Loungani (2017)
  - Job finding rates decrease (a lot) equally across households
  - Job separation rates decrease (a bit), higher elasticity for high-x households
    + Functional forms: additive fall in separation rates in recession delivers the pattern

Unemployment Steady State



Unemployment **Business Cycle: Okun's law**



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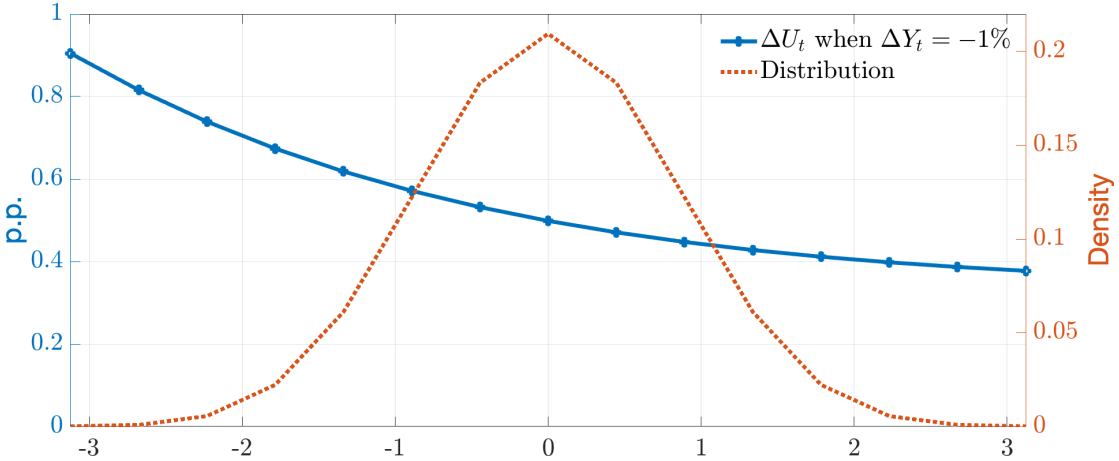


Investigating the Calibration **Household responses**

- Marginal propensities to consume (mpc)

  Parker, Souleles, Johnson, and McClelland (2013), Kaplan and Violante (2014), ...
  Saporta-Eksten (2014), Ganong and Noel (2019)

  – Compute mpc out of a $500 rebate: average quarterly mpc at 0.30

  – Decline with wealth: from 0.92 to 0.03 from 1st to 5th wealth quintile

Wealth quintile	1	2	3	4	5
Wealth share	0.00	0.01	0.02	0.12	0.85
mpc	0.92	0.34	0.14	0.07	0.03

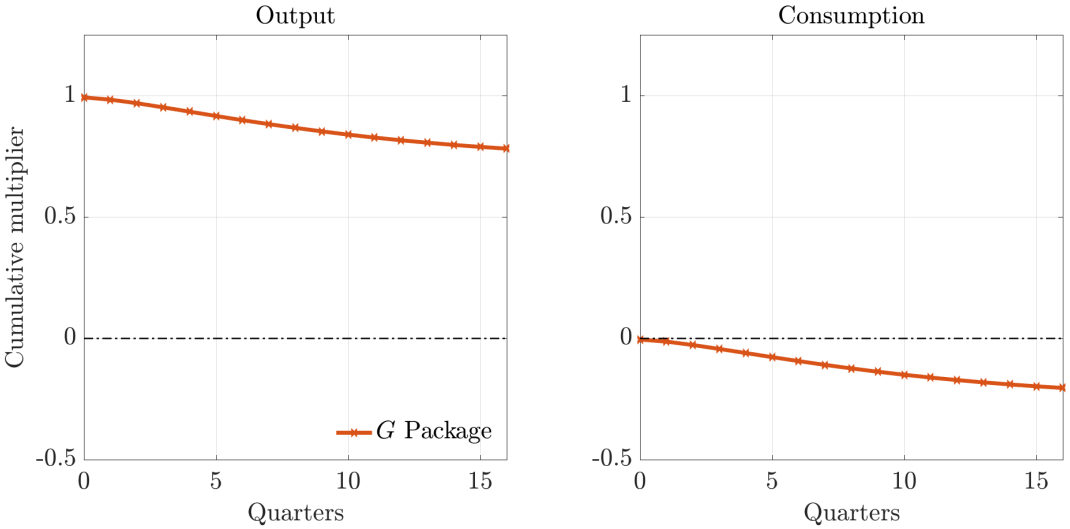
  – Larger for unemployed at 0.55

Spending, Checks, and Consumption Taxes

Three Fiscal Packages

- Three packages of equivalent cost
  - Government spending shock with persistence $\rho_w = 0.75$
  - Consumption tax cut, same size & persistence
  - Lump-sum check, same size & persistence
- Three sets of results
  - Sticky wages/homogeneous intensive labor supply
  - Robustness: Sticky prices & extensive labor supply at the household level
  - Robustness: Sticky wages, higher risk aversion: $\sigma = 2$, $\varphi^{-1} = 1/3$

Three Fiscal Packages



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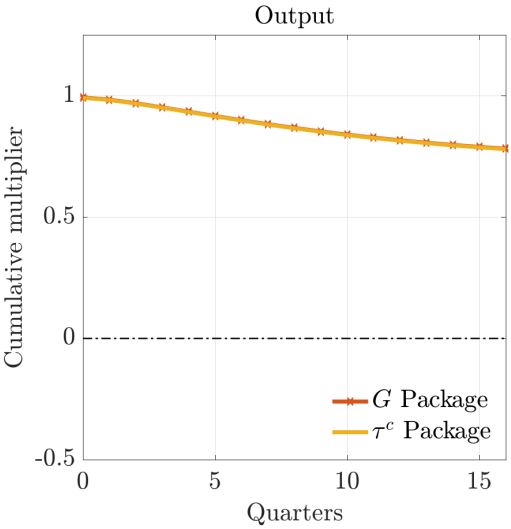

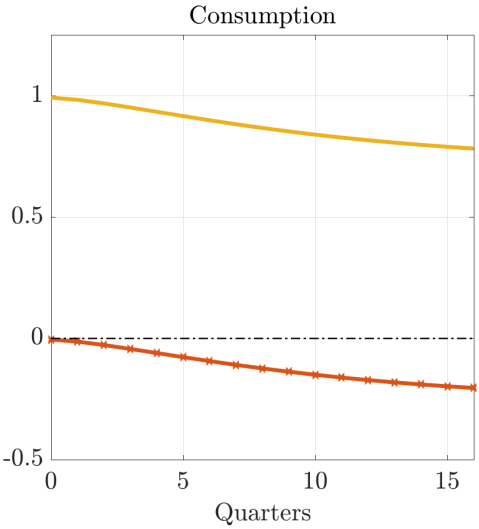

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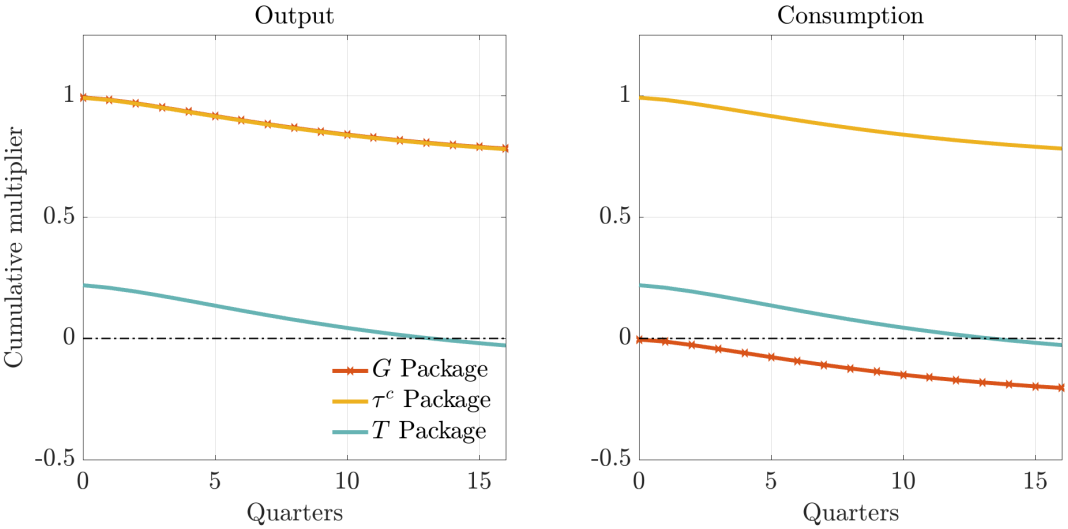
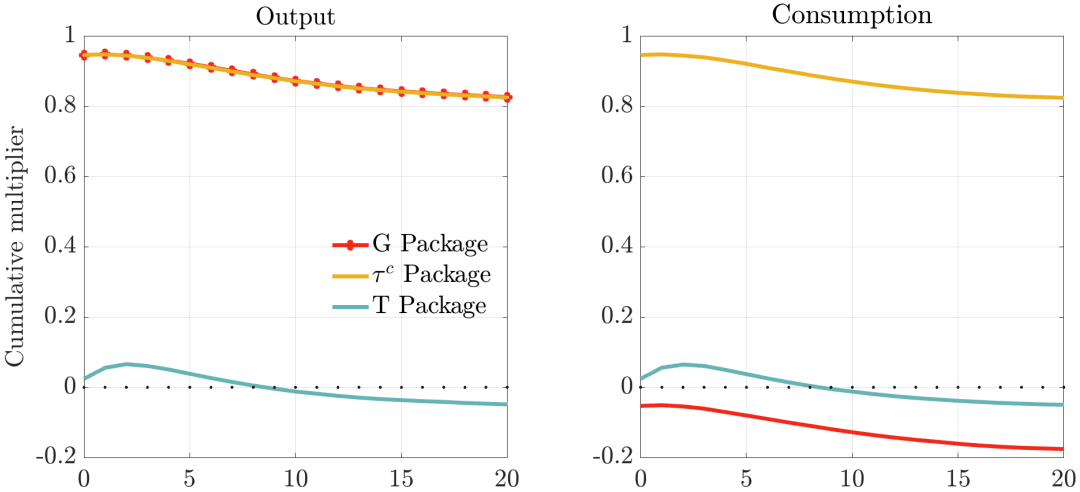


Three Packages Robustness: Extensive Labor & Sticky Prices

Three Packages **Robustness: Extensive Labor & Sticky Prices**



Three Packages Robustness: Higher Risk Aversion

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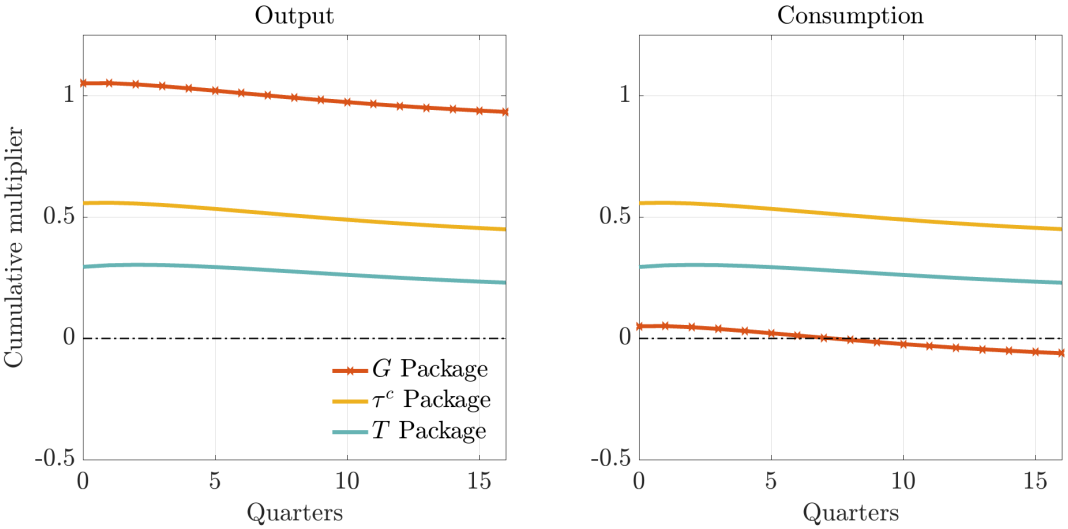

Recession

Recession Demand Shock

- Recession induced by a negative demand shock: $(1-\omega_t)u(c_t, n_t)$
  - ω_0 such that $\Delta Y_t = -0.1\%$ on impact
  - Reverts to $\omega = 0$ with persistence $\rho_\omega = 0.75$ at the quarterly level
- Unexpected, transitory, perfect foresight: a 'MIT' shock

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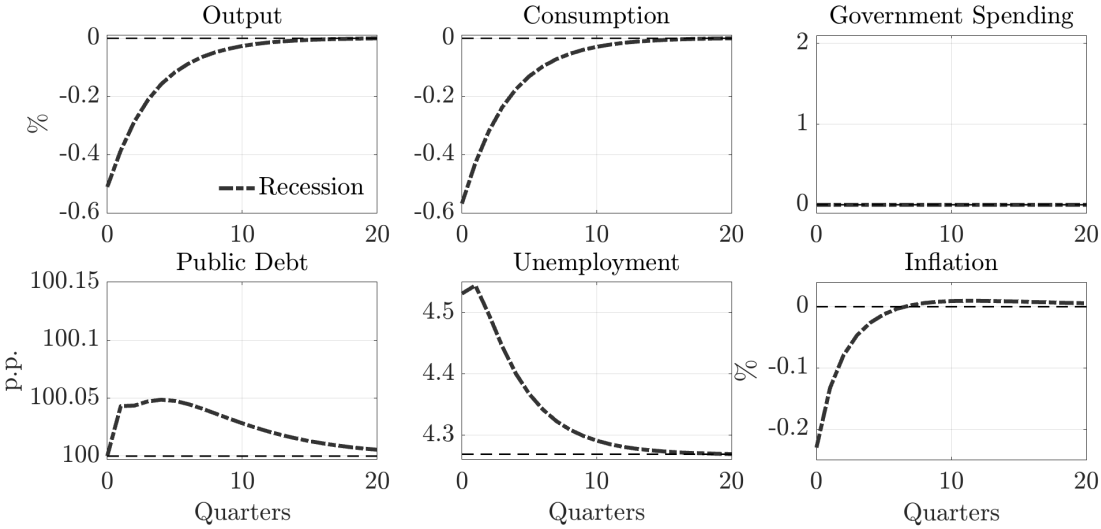
- Unexpected, transitory, perfect foresight: a 'MIT' shock
- Three packages of equal fiscal cost: G, τ^c, T
  - G package: Reduce the output loss by $\approx 40\%$ on impact, persistence ρ_w

Recession Stabilization



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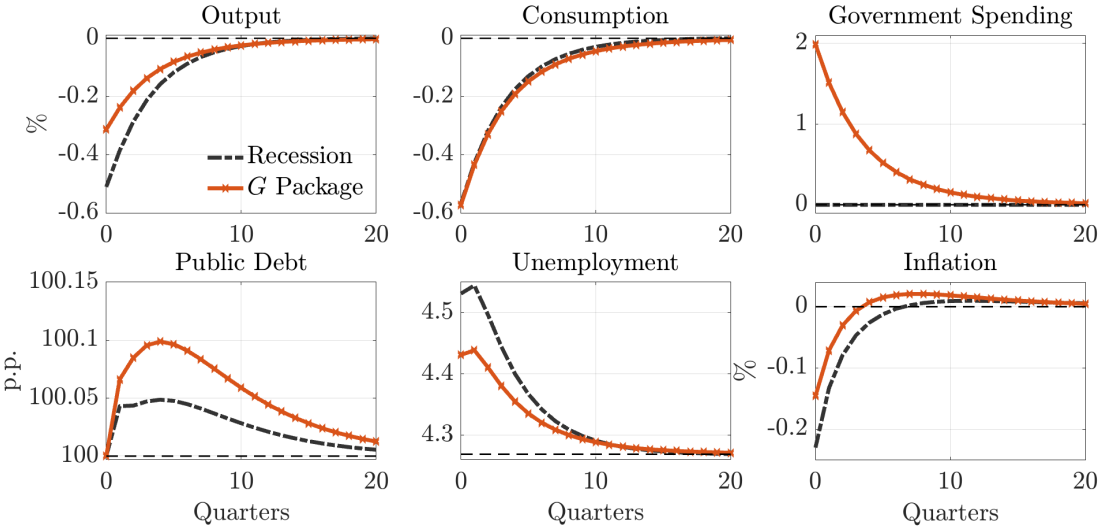

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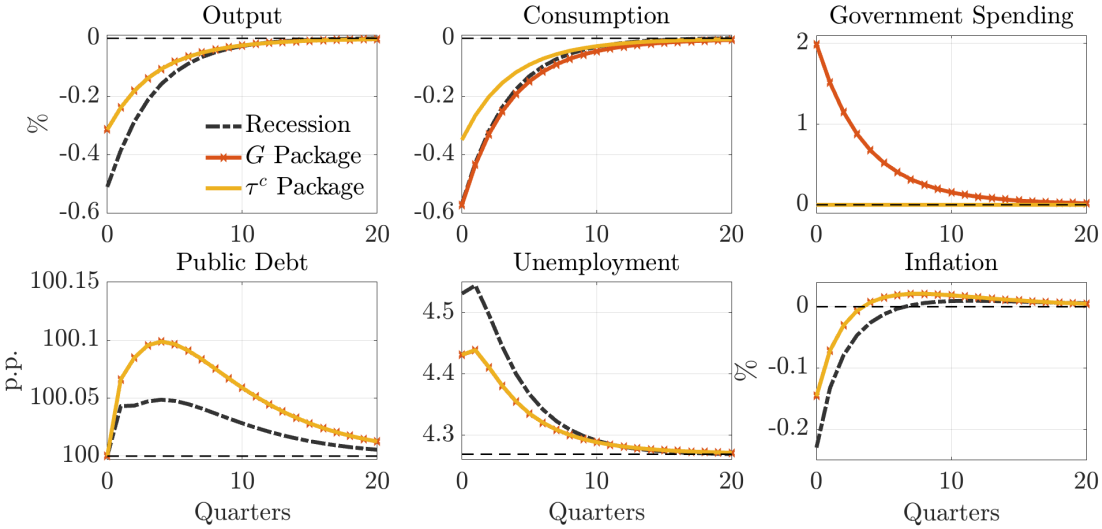

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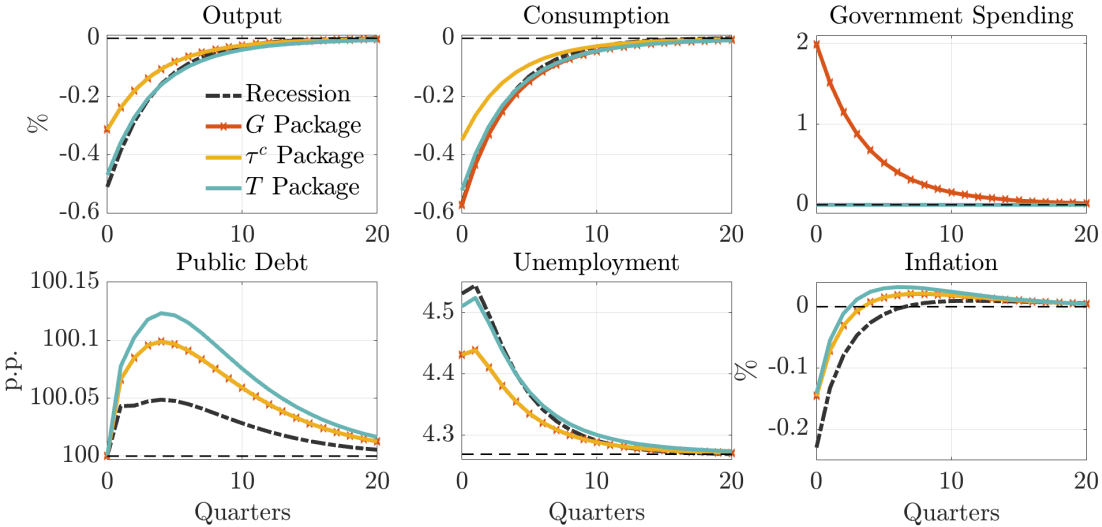
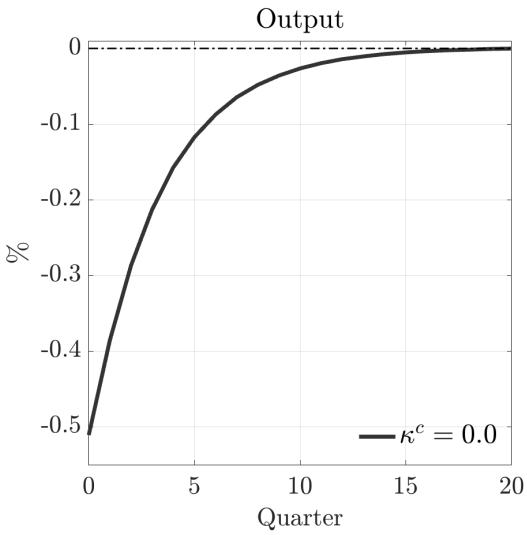


Recession Automatic Stabilizer

- Shall we use consumption taxes as an automatic stabilizer?
- Define a systematic rule: $\tau_t^c = \tau_c + \kappa^c \Delta Y_t$
  - κ^c captures the response of τ^c to the cycle
- Vary $\kappa^c \in \{0, 0.5, 1, 1.5, 2\}$

Recession Automatic Stabilizer



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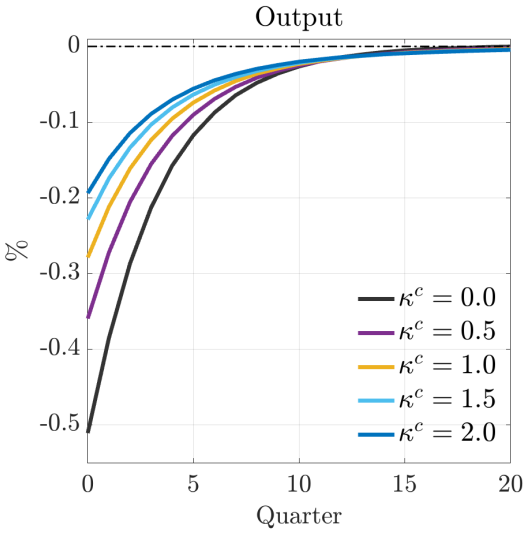

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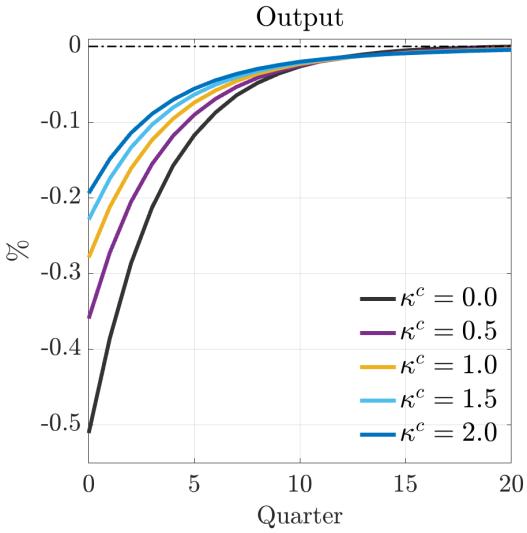

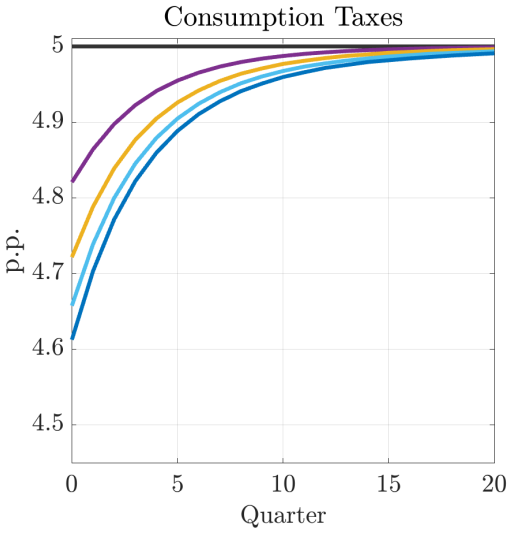

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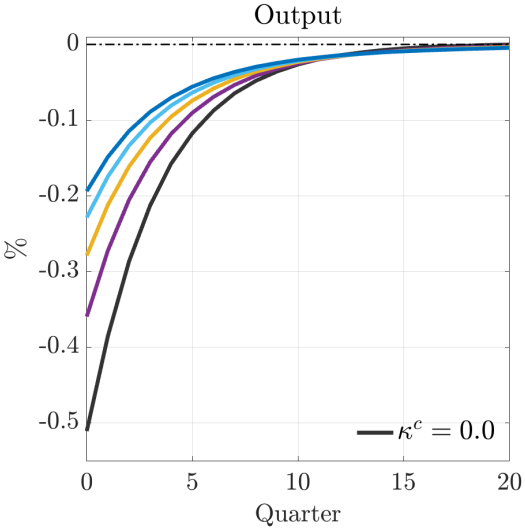
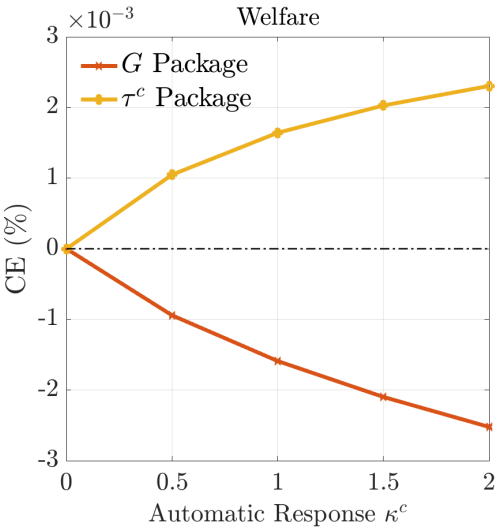




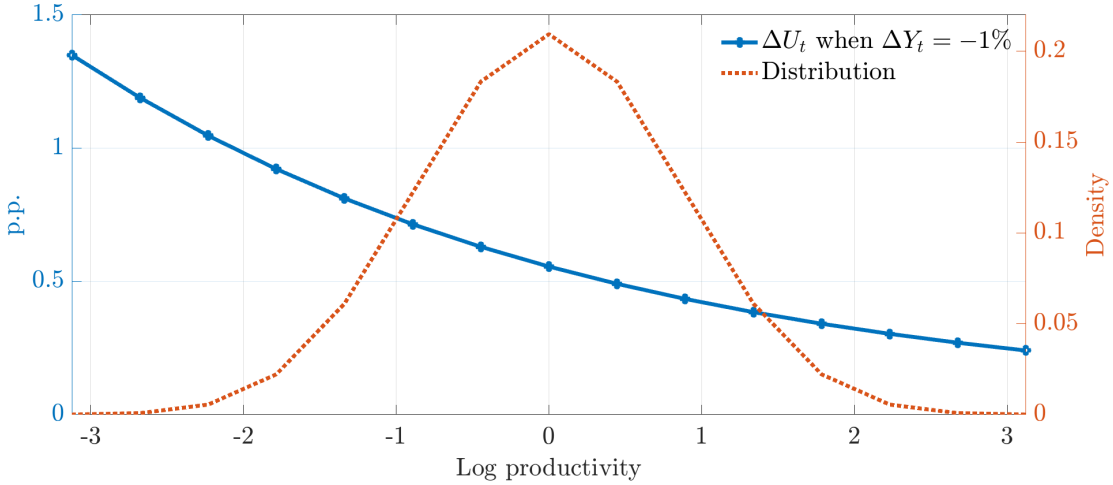
Distributional Costs **Okun's Law**

- New calibration with higher unemployment risk for the poor in recessions
  - Higher unemployment persistence
  - Higher gradient of separation rate with productivity

$\Rightarrow$ Slightly larger unemployment response overall

- Compare welfare gains of τ^c Package to the benchmark calibration

Distributional Costs **Okun's Law**



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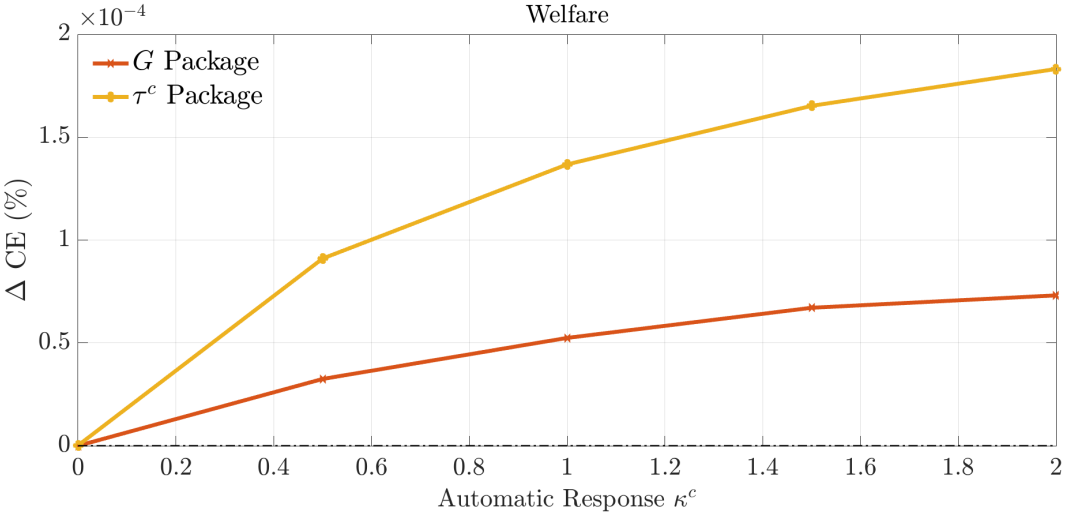

Fiscal Stabilization Going Forward

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- Valuation of G
- Going further: baskets of goods? durables?

Alternative Fiscal Instruments

Fiscal Stabilization Packages **Alternative Instruments**

- Two fiscal stabilization packages
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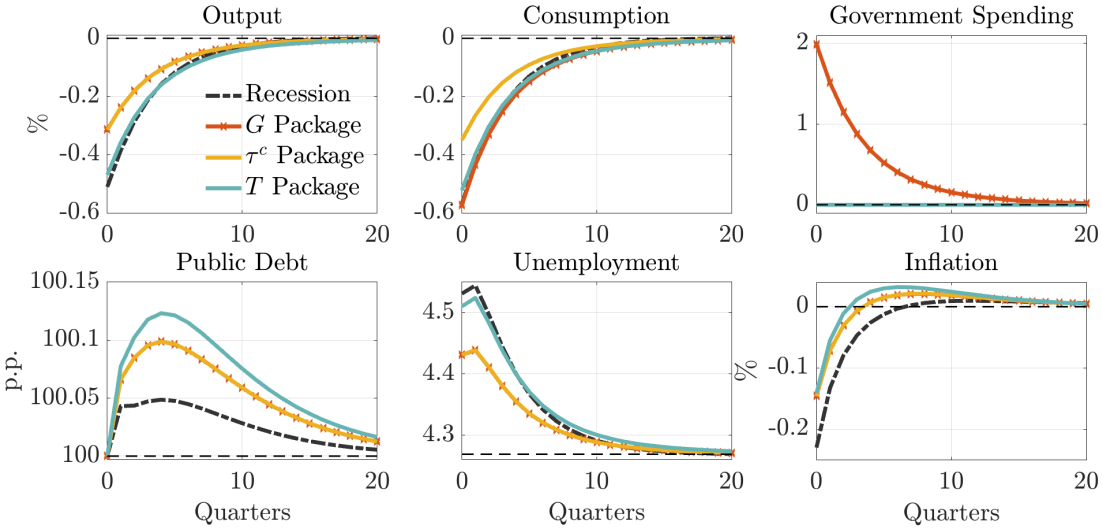
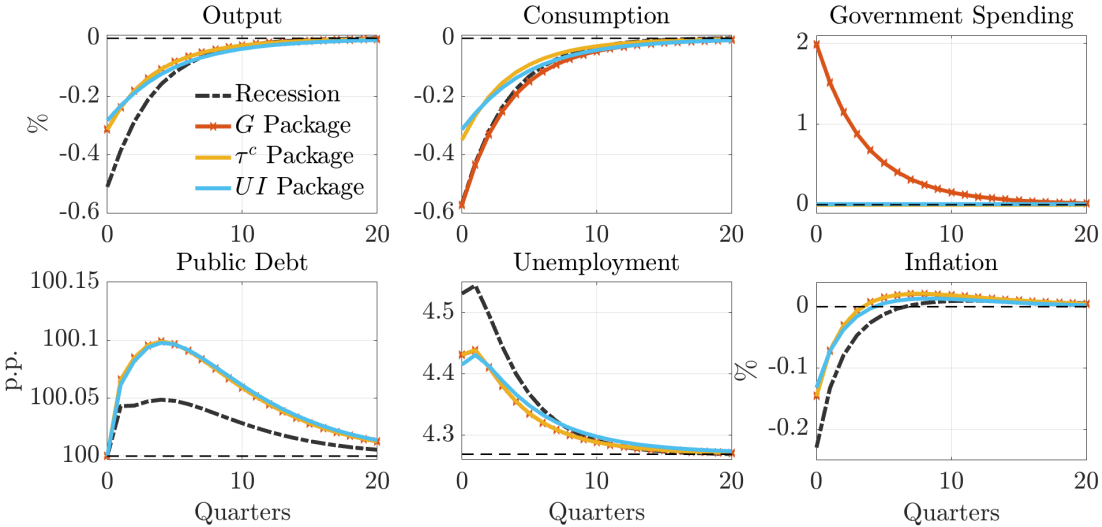
- Similar exercise: same fiscal cost
  - Only UI today!

Recession Stabilization with UI



Recession Stabilization with UI



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Thank you!

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Fiscal Rule

- Public debt adjusts as a function of Φ_D

$$D_{t+1} = (1-\phi_D)D + \phi_D\left(\hat{G}_t - \tau^k r_t A_t - \tau_t^c C_t - \mathcal{R}_t^\ell\right), \text{ where}$$

  - $\hat{G}_t$ captures total government expenditures, including debt repayments

    $\hat{G}_t = G_t + T_t + \mathcal{U}_t + (1+r_t)D_t$

  - $\mathcal{R}_t^\ell$ captures fiscal revenues at steady-state labor tax schedule

    $\mathcal{R}_t^\ell = w_t L_t - \lambda \int (w_t x h_t(a,x,\eta,\beta))^{1-\tau^\ell} d\mu_t(a,x,\eta,\beta)$

Return

Steady State Households

- Quarterly model calibrated to liquid wealth

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- Stochastic $\beta \in \{\bar{\beta} - \Delta,\ \bar{\beta},\ \bar{\beta} + \Delta\}$, duration of 50 years
  Krusell and Smith (1998)
  - $\bar{\beta}$ s.t. $r \equiv 3.5\%$ annually
  - Δ s.t. top-quintile liquid wealth $\approx 90\%$ (SCF)

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  - B to match employment $\approx 78\%$
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  - ρ_h to match average annual labor elasticity of ≈ 0.3
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- Productivity $(\rho_x, \sigma_x) = (0.989, 0.287)$
  Chang and Kim (2007)

Return

Steady State **Firm and government**

- Technology: $\varepsilon = 7$, $\Theta = 200 \rightsquigarrow$ Phillips curve slope $\varepsilon/\Theta = 0.035$

  Galí and Gertler (1999)

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    Chen, Imrohoroglu, and Imrohoroglu (2007), Heathcote, Storesletten, and Violante (2017), Ferriere, Grübener, Navarro, and Vardishvili (2023)
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    $\chi = 0.15$ to match $C_u/C_e \approx 75\%$

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  - Automatic responses of inflation and debt: $\Phi_\Pi = 1.5$, $\Phi_D = 0.75$

Return

Steady State **Unemployment**

- Job finding rates and separation rates across hourly wage distribution

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  - Monthly finding rate of $0.32 \Rightarrow \pi_\eta(\ell|u) = 0.691$

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  - Monthly separation rates of $\approx 1.4\%$ and 0.7% below and above median, respectively

  $\Rightarrow$ $\pi_\eta(u|\ell, x) = \phi_0 x^{\phi_1}$, with $\phi_0 = 0.029$ and $\phi_1 = -0.446$

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- Average unemployment rate at 4.3% with unequal incidence in the distribution

Return

Unemployment and the Business Cycle

- Okun's law type of relation between output and unemployment
  - Okun coefficient $c^{OK} = 0.5$

    Ball, Leigh, and Loungani (2017)

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  - *Elasticity* of separation rates to aggregate unemployment larger for above-median workers
    Mueller (2017)
  - + Homogeneous additive increase in separation rates

Unemployment and the Business Cycle **Okun's law**

- Finding and separation rates distribution depend on U_t Mueller (2017)
  - Finding rate elasticity decreases homogeneously with ΔY_t

$$\log \pi_{\eta,t}(\ell|u,x) = \log \pi_\eta(\ell|u) - \log\left(1 - \bar{\phi}_e \Delta Y_t\right)$$

  - Separation rate elasticity increases with ΔY_t

$$\pi_{\eta,t}(u|\ell,x) = \pi_\eta(u|\ell,x) - \bar{\phi}_u \Delta Y_t x^{-\phi_{u,x}}$$

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  - Joint calibration:
    + $\bar{\phi}_e$ s.t. finding elasticity to $U \approx -0.6$
    + $\phi_{u,x} = 0$ elasticity of separation rates larger for above-median workers
    + $\bar{\phi}_u = 0.33$ to get $c_{OK} = 0.5$

Return

Deeper Recessions **Bigger Fiscal Packages**

- Consider a recession of about 1% on impact – compared to 12bp on impact in the baseline
- Implement fiscal packages costing $\$1500$ per household

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- TT Package in the first quarter: equal to $\$1100$ per month for the bottom 5%, $\$500$ per month for the 5-15%
- TC Package in the first quarter: equal to $\$1100$ per month for the bottom 5%, $\$500$ per month for the 5-15%
- UI Package in the first quarter: equal to $\$2800$ per month for all unemployed

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- Multipliers are similar to the baseline

Return